

Sparky Build Shopping List

Electronics needed to bring the physical Sparky tutor to life (enclosure excluded - 3D printed separately)

Checked across the approved vendor list, South Africa, 2026-09-20

This list covers everything needed to make Sparky fully functional (mic, speaker, expressive LED face, accent light, push-to-talk button, and prototyping supplies) on top of the ESP32-WROOM board already on hand. The enclosure is excluded, since it's being 3D printed separately.

Checked across five vendors from the approved vendor list: Micro Robotics, Mantech, Communica, DIY Electronics, and Takealot. Two items (the microphone and amplifier) were out of stock at Micro Robotics at check time; Communica stocks both (prices confirmed directly from their live listing), consolidating what would otherwise be a split order.

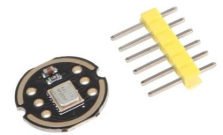
Touch response (tapping Sparky's head for a reaction) is free - the ESP32-WROOM has built-in capacitive touch GPIOs, nothing to buy.

USB-C cable: not included - already owned from earlier ESP32 testing, no need to buy another.

Power bank: not needed - the user already owns one (believed ~10,000mAh). It has the same low-current auto-shutoff behavior discussed earlier for a purchased bank, but a workaround (keep-alive trickle load) is already planned, so no purchase is required.

1. INMP441 I2S Microphone Module

Digital I2S microphone. Captures the child's spoken question and streams it to the ESP32 over I2S - no separate ADC needed. Micro Robotics' listing (SEN0327) was out of stock, then Communica's exact-chip listing (R65.00) was the recommendation - but the user confirmed Communica is now SOLD OUT too. RECOMMENDED: DIY Electronics stocks the exact INMP441 directly (also the vendor with the standing NWU-invoice process) - doesn't label items 'in stock' but shows 'ships within 1 day', a strong signal it's actually available. Price still needs confirming on the live page. Alternative: Takealot lists a 5-pack (INMP441 Omnidirectional Module, 5 units) that IS available, but ships 14-16 working days from a global/international seller - much slower, worth knowing before picking it for time-sensitive ordering. Mantech lists an INMP441 board too, but their listing title says 'I2C Interface', which conflicts with the chip's actual I2S interface - confirm with Nasen before trusting that one. NOTE: 'ships within 1 day' is dispatch time only - DIY Electronics' own delivery page adds a further 2-4 working days standard courier transit on top, so total arrival is more like 3-5 working days. Free shipping only above ~R1000-R1250 (site shows both figures); a small order like this won't qualify, so a shipping fee applies (exact amount calculated at checkout).



Price: R119.00 | Supplier: DIY Electronics (proposed) | Stock: Ships within 1 day (DIY Electronics) - likely available

<https://www.diyelectronics.co.za/store/audio/5380-inmp441-omnidirectional-mems-microphone-module.html>

2. MAX98357A I2S Amplifier Module

I2S class-D amplifier. Converts Sparky's synthesized speech straight into an analog signal driving the speaker - no separate DAC needed. Micro Robotics' listing (DFR0954) was out of stock; Communica's MAX98357A-based module (HKD MAX98357) is the replacement and is confirmed in stock.

Price: R75.00 | Supplier: Communica | Stock: In stock (confirmed 2026-09-20)

<https://www.communica.co.za/products/hkd-max98357-i2s-3w-class-d-amp>



3. Enclosed Speaker 3W 8 Ohm (FIT0502)

Small enclosed speaker that plays Sparky's spoken answers. Pairs with the MAX98357A amp above (same DFRobot family, matching connector).

Price: R89.70 | Supplier: Micro Robotics | Stock: In stock - Centurion & Stellenbosch (confirmed 2026-09-20)

<https://www.robotics.org.za/FIT0502>

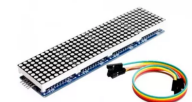


4. MAX7219 Dot Matrix Display 32x8 - Blue

Four chained 8x8 LED matrices forming Sparky's face - drives the idle/happy/thinking/frowning animations discussed for the interactivity design. Also available in red or green if you'd rather match a different pixel color; purely cosmetic.

Price: R110.40 | Supplier: Micro Robotics | Stock: In stock - Centurion & Stellenbosch (confirmed 2026-09-20)

<https://www.robotics.org.za/MAX7219-DOT-BLU>



5. LED 5mm - Clear White (10-pack, LED-05-WHI)

Accent light for the lightning-bolt logo - pulses while Sparky is thinking, steady while idle. Swapped from the original WS2812B addressable RGB LED to this plain through-hole LED: cheaper (R6.90 vs R18.40+), and has real legs that plug straight into the breadboard (the WS2812B options were either SMD pads needing solder, or a pricier TH version). Trade-off: no more color-changing capability, only on/off/pulse in white - fine, since that's all the current design actually uses. 3.2V forward voltage, 10mA recommended (per packaging) - not enough headroom off the 3.3V rail for a resistor to reliably limit current, so this runs off the 5V rail instead with the ESP32 pin acting as a low-side switch (GPIO LOW = LED on, HIGH = LED off - inverted from the usual sourcing setup). Needs a series resistor - user already has an assortment on hand, so none purchased here. Target value: $(5V - 3.2V) / 0.010A = \sim 180 \text{ ohm}$; anything in the 150-330 ohm range works fine for an indicator LED like this (220 ohm is the safe default if unsure which to grab).

Price: R6.90 | Supplier: Micro Robotics | Stock: In stock (confirmed 2026-09-21)

<https://www.robotics.org.za/LED-05-WHI>



LED diameter 5 mm
Light colour Cool white
Forward voltage 3.2 V
Forward current 10 mA recommended

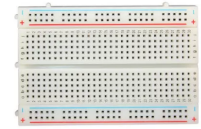
6. Breadboard, Half Size, 400 Points (BREAD-400)

Solderless prototyping board - wire up and test the mic, amp, matrix, and accent LED before committing to a final soldered build inside the 3D-printed shell.

Price: R20.70 | Supplier: Micro Robotics | Stock: In stock (confirmed 2026-09-20)

<https://www.robotics.org.za/BREAD-400>

BREADBOARD HALF SIZE



400 TIE POINTS

SIZE 55 X 83

7. Jumper Wires, Male-Female, 50-pack (JUMP-MF)

Connects breakout board pins (mic, amp, matrix, LED) to the ESP32's GPIO headers on the breadboard. Micro Robotics confirmed out of stock (user checked - no other M-F variant in stock there either). RECOMMENDED: DIY Electronics' Male to Female Breadboard Jumpers (40pcs, 21cm) - shows 'ships within 1 day' rather than an explicit stock label, a strong signal it's actually available. See the microphone entry above for the shipping-time and free-shipping-threshold caveat that applies to all DIY Electronics items in this list.

Price: R19.95 | Supplier: DIY Electronics (proposed) | Stock: Ships within 1 day (DIY Electronics) - likely available

<https://www.diyelectronics.co.za/store/connectors-wiring/258-male-to-female-breadboard-jumpers-40pcs-21cm.html>



8. Premium Jumper Wires, Male-Male, 20-pack

Male-male jumpers for breadboard-internal connections (e.g. power rail to rail) that the male-female pack above doesn't cover. Micro Robotics confirmed out of stock. RECOMMENDED: DIY Electronics' Male to Male (Dupont) Breadboard Jumpers (40pcs, 21cm) - same 'ships within 1 day' signal as the M-F pack above, and the same shipping-time/free-shipping caveat noted on the microphone entry.

Price: R16.00 | Supplier: DIY Electronics (proposed) | Stock: Ships within 1 day (DIY Electronics) - likely available

<https://www.diyelectronics.co.za/store/connectors-wiring/1168-dupont-jumper-cables-male-to-male-40pcs-21cm.html>



9. Push-to-Talk Button - 30mm Arcade Button (SANWA style, Blue)

Physical push-to-talk trigger - a learner holds this down to speak and releases to send the question. A big 30mm arcade button was chosen over a small 6mm tactile switch since it's much easier for a child to find and hold comfortably, and more durable under repeated use. Same vendor as the mic and both jumper packs, so no new shipping charge. Blue matches Sparky's brand palette better than the originally-considered purple/pink variant. SKU: 9SWJSA30MMBLU. Wiring: one leg to GPIO 14, the other to GND, using the ESP32's internal pull-up (no external resistor needed) - pressed reads LOW, released reads HIGH.

Price: R28.00 | Supplier: DIY Electronics (proposed) | Stock: In stock (confirmed 2026-09-20)

<https://www.diyelectronics.co.za/store/buttons-switches/5607-blue-arcade-push-button-30mm-sanwa-style.html>



10. Momentary Tactile Switch, 8-pack (TACC-66) - for breadboard testing only

NOT the final Sparky button - this is a cheap, genuinely breadboard-pin compatible tactile switch, bought purely to test the push-to-talk wiring and firmware logic on the breadboard without soldering or modifying the real 30mm arcade button. The arcade button uses 2.8mm spade terminals (not breadboard-pins), so testing with it directly would mean soldering or crimping it before you've even confirmed the circuit works. Use one of these for all breadboard testing, then swap in the real arcade button (solder or spade-connect it, your choice - not yet decided, doesn't matter until final assembly) once you're building the actual enclosure.



Price: R10.00 | Supplier: Micro Robotics | Stock: In stock - Centurion & Stellenbosch (confirmed 2026-09-22)

<https://www.robotics.org.za/TACC-66>

Summary Table

Name	Supplier	In Stock?	Description	Qty	Price	Link
INMP441 I2S Microphone Module	DIY Electronics (proposed)	Ships within 1 day (DIY Electronics) - likely available	Digital I2S microphone.	1	R119.00	Link
MAX98357A I2S Amplifier Module	Communica	In stock (confirmed 2026-09-20)	I2S class-D amplifier.	1	R75.00	Link
Enclosed Speaker 3W 8 Ohm (FIT0502)	Micro Robotics	In stock - Centurion & Stellenbosch (confirmed 2026-09-20)	Small enclosed speaker that plays Sparky's spoken answers.	1	R89.70	Link
MAX7219 Dot Matrix Display 32x8 - Blue	Micro Robotics	In stock - Centurion & Stellenbosch (confirmed 2026-09-20)	Four chained 8x8 LED matrices forming Sparky's face - drives the idle/happy/thinking/frowning animations discussed for the interactivity design.	1	R110.40	Link
LED 5mm - Clear White (10-pack, LED-05-WHI)	Micro Robotics	In stock (confirmed 2026-09-21)	Accent light for the lightning-bolt logo - pulses while Sparky is thinking, steady while idle.	1	R6.90	Link
Breadboard, Half Size, 400 Points (BREAD-400)	Micro Robotics	In stock (confirmed 2026-09-20)	Solderless prototyping board - wire up and test the mic, amp, matrix, and accent LED before committing to a final soldered build inside the 3D-printed shell..	1	R20.70	Link
Jumper Wires, Male-Female, 50-pack (JUMP-MF)	DIY Electronics (proposed)	Ships within 1 day (DIY Electronics) - likely available	Connects breakout board pins (mic, amp, matrix, LED) to the ESP32's GPIO headers on the breadboard.	1	R19.95	Link
Premium Jumper Wires, Male-Male, 20-pack	DIY Electronics (proposed)	Ships within 1 day (DIY Electronics) - likely available	Male-male jumpers for breadboard-internal connections (e.g.	1	R16.00	Link
Push-to-Talk Button - 30mm Arcade Button (SANWA style, Blue)	DIY Electronics (proposed)	In stock (confirmed 2026-09-20)	Physical push-to-talk trigger - a learner holds this down to speak and releases to send the question.	1	R28.00	Link
Momentary Tactile Switch, 8-pack (TACC-66) - for breadboard testing only	Micro Robotics	In stock - Centurion & Stellenbosch (confirmed 2026-09-22)	NOT the final Sparky button - this is a cheap, genuinely breadboard-pin compatible tactile switch, bought purely to test the push-to-talk wiring and firmware logic on the breadboard without soldering or modifying the real 30mm arcade button.	1	R10.00	Link

Stock statuses below are as confirmed directly by the user checking each vendor's live site on 2026-09-20. Three items (mic, both jumper packs) had their original recommended source go out of stock and now point to a confirmed replacement at DIY Electronics instead. All 10 items now have confirmed prices. Item 10 (tactile switch) is a breadboard testing aid, not part of the final device. No resistor purchase needed for the accent LED - the user has an assortment on hand. The USB-C cable and the power bank are not included - already owned.

Subtotal, all 10 items at firmly confirmed prices: R495.65

No resistor purchase needed - user has an assortment on hand (~180-220 ohm for the LED)

Plus shipping fees across 3 vendors (Micro Robotics, Communica, DIY Electronics) - not yet included

No power bank needed - user already owns one (see note above)

Parts-only total: R495.65, before shipping